BUSINESS REQUIREMENTS DOCUMENT

Customer Churn Prediction & Retention Strategy

TelecomX Analytics Initiative

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1. Executive Summary

TelecomX, a mid-sized telecom provider serving approximately 7,000 active customers, is experiencing a monthly customer churn rate of 26% — significantly above the industry benchmark of 15–18%. The absence of a data-driven early-warning system means the Customer Retention team relies on reactive outreach, resulting in wasted campaign spend on low-risk customers while high-risk customers leave unnoticed.

This Business Requirements Document defines the scope, objectives, stakeholders, functional requirements, and success criteria for the Customer Churn Prediction & Retention Strategy initiative. The project will deliver:

- A predictive churn model (logistic regression) identifying customers with >70% churn probability
- Identification of the top 5 behavioral and contractual drivers of churn
- An interactive Tableau Retention Dashboard for daily use by the Customer Success team
- Three data-backed business recommendations to reduce monthly churn by 5–8 percentage points

Business case summary

Current monthly churn rate: 26% | Industry benchmark: 15–18% | Gap: ~8–11 percentage points
Estimated cost per churned customer: $240 annual revenue lost + $65 re-acquisition cost = $305 total
High-risk customers identified per month (est.): ~380
Projected annual revenue at risk: $380 x $240 = $91,200
Target: Reduce churn by 5% via targeted retention => ~$45,600 annual revenue protected

2. Project Background & Problem Statement

2.1 Business Context

TelecomX operates in a highly competitive telecom market where customer acquisition costs have risen 23% year-over-year. Retaining an existing customer is estimated to be 5–7x cheaper than acquiring a new one. Despite this, the business currently has no systematic method to identify which customers are likely to leave before they submit a cancellation request.

The Customer Success and Marketing teams currently run monthly retention campaigns targeting 2,000+ customers based on broad demographic segments, resulting in low conversion rates and high operational cost per retained customer.

2.2 Problem Statement

Core business problem

TelecomX lacks a predictive capability to identify at-risk customers early enough to intervene.
Retention campaigns are reactive and broadly targeted, leading to:
- Budget waste: ~60% of campaign spend targets low-risk customers who would not have churned
- Missed interventions: High-risk customers leave before outreach occurs

- No insight into root causes: Teams cannot identify which products, contracts, or service issues drive churn

2.3 Business Opportunity

If TelecomX can predict churn 30 days in advance with >75% accuracy, the Customer Success team can proactively target high-risk customers with personalized retention offers, potentially reducing monthly churn by 5–8 percentage points and protecting an estimated $45,600–$73,000 in annual recurring revenue.

3. Project Objectives & Goals

3.1 Primary Objectives

- **OBJ-01:** Build a logistic regression model that predicts customer churn with >75% accuracy on the TelecomX customer dataset
- **OBJ-02:** Identify and rank the top 5 drivers of customer churn from behavioral, contractual, and service usage data
- **OBJ-03:** Develop an interactive Tableau Retention Dashboard with 4 views: churn overview, risk segmentation, top churn drivers, and high-risk customer action table
- **OBJ-04:** Deliver three actionable business recommendations with estimated revenue impact for presentation to the retention management team
- **OBJ-05:** Document all data quality findings, assumptions, and model limitations in a transparent, stakeholder-ready format

3.2 Business Goals (Non-Technical)

Business goal	Expected outcome
Proactive retention outreach	Identify high-risk customers 30 days before predicted churn
Campaign efficiency	Reduce broad targeting; focus spend on top 380 high-risk customers
Revenue protection	Protect ~$45,600–$73,000 annual recurring revenue
Insight-driven strategy	Equip leadership with data-backed retention policy decisions

4. Project Scope

4.1 In Scope

- Exploratory Data Analysis (EDA) on the IBM Telco Customer Churn dataset (7,043 records, 21 features)
- Data cleaning, preprocessing, and feature engineering for model input
- Logistic regression churn prediction model with precision, recall, and accuracy metrics
- Customer risk segmentation: High Risk, Medium Risk, Low Risk
- Interactive Tableau Public dashboard (4 views) published with a shareable link
- Business recommendations report (3 recommendations with estimated ROI)
- GitHub repository with structured folders: /data, /notebooks, /docs, /dashboard

4.2 Out of Scope

- Real-time integration with CRM or billing systems
- Deployment of a live API or automated model pipeline
- Deep learning or ensemble model approaches (XGBoost, Random Forest) — may be explored in Phase 2
- Customer interview research or primary data collection
- A/B testing of retention campaign strategies

4.3 Assumptions

- The IBM Telco Customer Churn dataset is an acceptable proxy for TelecomX customer data for the purposes of this project
- Historical churn patterns in the dataset are representative of future behavior
- Revenue impact estimates are based on an assumed average annual revenue of $240 per customer and $65 re-acquisition cost
- Model will be used for monthly batch scoring, not real-time prediction

4.4 Constraints

- All tools used must be free or open-source (Python, Tableau Public, GitHub)
- Project must be completable within 3–4 weeks as a solo initiative
- No access to live customer or transactional databases — analysis limited to static dataset

5. Stakeholder Register

The following stakeholders have been identified for this initiative. In a real-world engagement, these individuals would be consulted during requirements gathering, review checkpoints, and final delivery.

Stakeholder	Role	Interest	Influence	Engagement
Retention Manager	Primary user of dashboard	Daily churn risk visibility	High	Consulted & Informed
Head of Marketing	Campaign strategy owner	ROI on retention spend	High	Consulted

Stakeholder	Role	Interest	Influence	Engagement
Data Analytics Team	Technical reviewers	Model accuracy, data quality	Medium	Accountable
Customer Success Reps	End users of risk table	Easy-to-act customer list	Low	Informed
Senior Leadership	Decision & budget approval	Revenue impact, ROI	High	Informed

6. Functional Requirements

6.1 Data Requirements

ID	Requirement	Acceptance Criteria
DR-01	Load IBM Telco dataset (7,043 records, 21 columns)	Dataset loads without errors; row/column count verified
DR-02	Identify and document all data quality issues	Data quality log created; missing values in TotalCharges resolved
DR-03	Encode categorical variables and scale numeric features	All features are numeric; no null values remain post-cleaning
DR-04	Create 3 customer risk segments from EDA	Segments defined with clear business rules and documented criteria

6.2 Model Requirements

ID	Requirement	Acceptance Criteria
MR-01	Build logistic regression classification model	Model trained on 80% of data; tested on 20% holdout
MR-02	Achieve minimum 75% model accuracy	Accuracy ≥ 75% on test set; confusion matrix documented
MR-03	Extract top 5 churn drivers from model coefficients	Feature importance table with business interpretation for each driver
MR-04	Export churn probability scores to CSV	Output CSV includes CustomerID, ChurnProbability, RiskSegment
MR-05	Translate model metrics into business language	Narrative summary: how many customers identified, estimated revenue saved

6.3 Dashboard Requirements

ID	Requirement	Acceptance Criteria
DB-01	Churn Overview view: churn rate %, total customers, revenue at risk, month-on-month trend	All 4 KPIs visible; values match Python model output
DB-02	Risk Segment Heatmap: customers by risk band x contract type and tenure	3 segments visible; drill-down by contract type functional
DB-03	Top Churn Drivers view: horizontal bar chart of top 5 drivers	Chart labelled with driver name and relative importance score
DB-04	High-Risk Action Table: sortable list of at-risk customers with recommended action per risk profile	Table sortable by churn probability; minimum 3 action types shown
DB-05	Dashboard published on Tableau Public with shareable URL	Link accessible without login; loads within 5 seconds

7. Key Performance Indicators (KPIs)

The following KPIs define both the model performance benchmarks and the business outcomes to be tracked via the Tableau dashboard.

KPI	Current Baseline	Target	Measurement Method
Monthly churn rate	26%	≤ 20% (within 6 months)	Churned customers / total customers
Model accuracy	N/A	≥ 75%	Scikit-learn accuracy_score on test set
Model precision (churn class)	N/A	≥ 70%	True positives / (true pos + false pos)
High-risk customers identified monthly	0 (no model)	~380 customers	Customers with churn probability > 70%
Revenue at risk (monthly)	$91,200/yr unmitigated	Reduce by $45,600/yr	High-risk count x $240 avg annual revenue
Retention campaign precision	~40% reach accuracy	≥ 70%	% of outreach targeting confirmed churners
Dashboard adoption	N/A (no dashboard)	Weekly use by retention team	Tableau Public view count

8. Data Dictionary

The following table documents the key fields in the IBM Telco Customer Churn dataset used for this project. Data quality issues are noted where applicable.

Field name	Data type	Description	Used in model	Notes
CustomerID	String	Unique customer identifier	No (ID only)	Used in output CSV
tenure	Integer	Months as a customer	Yes	Key churn driver
Contract	Categorical	Month-to-month / One year / Two year	Yes (encoded)	Highest churn predictor
MonthlyCharges	Float	Current monthly bill amount	Yes	Scaled for model
TotalCharges	String (issue)	Total billed to date	Yes (cleaned)	DATA QUALITY: 11 blank values — converted to float, blanks filled with 0
InternetService	Categorical	DSL / Fiber optic / No	Yes (encoded)	Fiber optic = higher churn
TechSupport	Categorical	Yes / No / No internet	Yes (encoded)	No support = higher churn
PaymentMethod	Categorical	4 payment types	Yes (encoded)	Electronic check = highest churn
Churn	Binary (Yes/No)	Target variable	Yes	Converted to 1/0 for model

9. Business Recommendations

Based on the churn driver analysis, three strategic recommendations are proposed. These will be validated against model output and presented to the Retention Manager and Marketing Head upon project completion.

Recommendation 1 — Contract upgrade incentive at month 3

Driver addressed: Month-to-month contracts have 3x higher churn rate than 2-year contracts

Action: Trigger an automated offer for a 1-year contract discount (10–15% off) at month 3, before churn risk peaks

Projected impact: If 20% of high-risk month-to-month customers convert, estimated annual revenue protected = $18,240

Recommendation 2 — Tech support bundling for fiber optic customers

Driver addressed: Fiber optic customers without TechSupport show 41% churn rate vs. 20% with support

Action: Offer complimentary 3-month TechSupport trial to fiber optic customers in their first 6 months

Projected impact: Reducing fiber churn by 5% protects approximately $14,400 in annual revenue

Recommendation 3 — Loyalty milestone program at 6-month and 12-month tenure

Driver addressed: Customers with tenure < 12 months account for 67% of all churn events

Action: Introduce loyalty rewards (bill credits, speed upgrades) triggered automatically at 6-month and 12-month milestones

Projected impact: Early tenure retention improvement of 3–5% could protect $21,600–$36,000 annually

10. Project Timeline

Phase	Duration	Key Deliverables	Milestone
Phase 1	Week 1 (Days 1–3)	BRD, stakeholder register, KPI definitions, success criteria	BRD sign-off
Phase 2	Week 1–2 (Days 4–10)	EDA notebook, data quality log, customer risk segments, churn driver findings	EDA complete
Phase 3	Week 2–3 (Days 11–17)	Trained logistic regression model, accuracy metrics, feature importance table, output CSV	Model validated
Phase 4	Week 3–4 (Days 18–24)	Tableau dashboard (4 views), Tableau Public link, dashboard user guide	Dashboard live
Phase 5	Week 4 (Days 25–28)	Business recommendations report, GitHub repo, final resume bullets	Project complete

11. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation
Model accuracy below 75% threshold	Medium	High	Tune hyperparameters; try feature selection; document final metrics transparently
Dataset not representative of real telecom customers	Medium	Medium	Document as assumption; note in BRD; recommend validation against real data in Phase 2

Risk	Likelihood	Impact	Mitigation
Tableau Public dashboard performance issues with 7,000 rows	Low	Low	Aggregate data at segment level; use extracts rather than live connections
Revenue impact estimates are approximations	High	Low	Clearly label all revenue figures as estimates; document assumptions used in calculation
Scope creep into Phase 2 ML or CRM integration	Medium	Medium	Strictly adhere to in-scope/out-of-scope definitions in Section 4; defer to Phase 2

12. Document Approval & Sign-Off

This BRD requires review and approval from key stakeholders before project execution commences. The approved version will be stored in the project GitHub repository under /docs/BRD.pdf.

Name	Role	Signature	Date
Shrusti Khadji	Business Analyst (Author)	_____	May 2025
[Retention Manager]	Primary Stakeholder	_____	
[Head of Marketing]	Sponsor	_____	
[Data Analytics Lead]	Technical Reviewer	_____	

Appendix A — Version History

Version	Date	Author	Changes
v1.0	May 2025	Shrusti Khadji	Initial release — all sections complete

Appendix B — Glossary

Term	Definition
BRD	Business Requirements Document — a formal document defining the business problem, objectives, scope, and requirements for a project
Churn	The event of a customer cancelling or not renewing their service subscription

Term	Definition
Churn rate	The percentage of customers who churned in a given time period out of total active customers
Logistic regression	A supervised machine learning algorithm that predicts the probability of a binary outcome (churn vs. no churn)
KPI	Key Performance Indicator — a measurable value used to evaluate the success of a business objective
EDA	Exploratory Data Analysis — the process of summarizing and visualizing data to discover patterns and anomalies before modelling
Feature importance	A measure of how much each variable contributes to a model's predictions
UAT	User Acceptance Testing — the process of verifying that a solution meets business requirements before sign-off
DXA	Document Transfer Accumulators — a unit of measurement used in OOXML (1440 DXA = 1 inch)